

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

MOHSIN IDNA HOSSAIN
AIUB, MATH-5

FINAL TERM

TOPIC NAME : _____

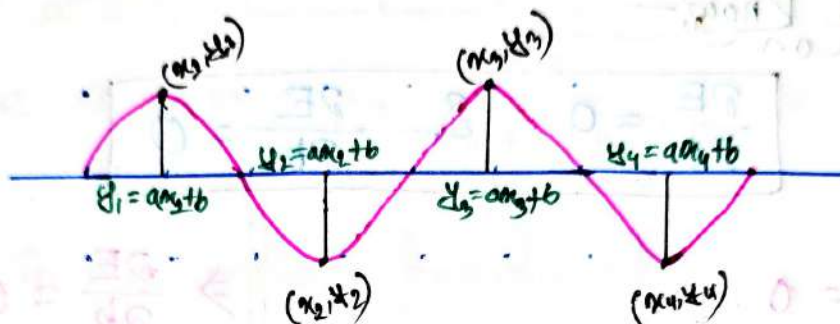
TIME : _____

DATE : / /

#Cense ~~HITTING~~
o o o o o
o o o o o
o o o o o

Curve Fitting

⇒ Fit a curve by a straight line.



$$E_1 = ax_1 + b - y_1$$

$$E_2 = ax_2 + b - y_2$$

$$\vdots$$

$$E_n = ax_n + b - y_n$$

So, $E = E_1 + E_2 + E_3 + \dots + E_n$

$$\therefore E = \sum_{i=1}^n E_i$$

Least Square Error Method

⇒ we know

$$E = E_1 + E_2 + \dots + E_n$$

Now $E = E_1^2 + E_2^2 + \dots + E_n^2$

$$\Rightarrow E = \sum_{i=1}^n E_i^2$$

$$\therefore E = \sum_{i=1}^n (ax_i + b - y_i)^2$$

Since,

If $E = 0$

Then $E^2 = 0$

$$\therefore E(a, b) = \sum_{i=1}^n (ax_i + b - y_i)^2$$

For Minimum

we know

$$\frac{\partial E}{\partial a} = 0 \quad \& \quad \frac{\partial E}{\partial b} = 0$$

$$\Rightarrow \frac{\partial E}{\partial a} = 0$$

$$\Rightarrow \frac{\partial}{\partial a} \left[\sum_{i=1}^n (ax_i + b - y_i)^2 \right] = 0$$

$$\Rightarrow 2 \sum_{i=1}^n (ax_i + b - y_i) \cdot x_i = 0$$

$$\Rightarrow \sum_{i=1}^n (ax_i^2 + bx_i - y_i x_i) = 0$$

$$\Rightarrow \sum_{i=1}^n ax_i^2 + \sum_{i=1}^n bx_i = \sum_{i=1}^n x_i y_i$$

$$\Rightarrow a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i = \sum_{i=1}^n x_i y_i$$

①

$$\Rightarrow \frac{\partial E}{\partial b} = 0$$

$$\Rightarrow \frac{\partial}{\partial b} \left[\sum_{i=1}^n (ax_i + b - y_i)^2 \right] = 0$$

$$\Rightarrow 2 \sum_{i=1}^n (ax_i + b - y_i) \cdot 1 = 0$$

$$\Rightarrow a \sum_{i=1}^n x_i + b \sum_{i=1}^n 1 = \sum_{i=1}^n y_i$$

$$\Rightarrow a \sum_{i=1}^n x_i + nb = \sum_{i=1}^n y_i$$

②

So The Normal equations are:-

$$\Rightarrow a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i = \sum_{i=1}^n x_i y_i$$

$$\Rightarrow a \sum_{i=1}^n x_i + nb = \sum_{i=1}^n y_i$$

Example 1:-
~~mon~~

⇒ Consider the data as:-

t (years)	2	4	6	8
P (lacs)	20.50	17.25	14.50	11.58

Using the equation $P = \frac{50}{a_2 + a_3 e^{t/4}}$

- (i) Convert in linearized form.
- (ii) Use least square method to find Normal equation
- (iii) Find value of $a_2 = 1$, a_3 to 3 decimal points.
- (iv) Estimate the value of a car of this type that is 10 years and the original price.
- (v) Use MATLAB function $a = \text{lsqcurvefit}(\text{fun}, a_0, X_{\text{data}}, Y_{\text{data}})$ for $P = \frac{a_1}{a_2 + a_3 e^{t/4}}$

Is given:-
~~mon~~

$$P = \frac{50}{a_2 + a_3 e^{t/4}}$$

$$\Rightarrow a_2 + a_3 e^{\frac{t}{4}} = \frac{50}{P}$$

$$\Rightarrow a_3 e^{\frac{t}{4}} = \frac{50}{P} - a_2$$

$$\Rightarrow \ln(a_3 e^{\frac{t}{4}}) = \ln\left(\frac{50}{p} - a_2\right) \quad \left\{ \begin{array}{l} \ln(ab) = \ln a + \ln b \\ \ln e = 1 \end{array} \right.$$

$$\Rightarrow \ln(a_3) + \ln e^{\frac{t}{4}} = \ln\left(\frac{50}{p} - a_2\right)$$

$$\Rightarrow \ln\left(\frac{50}{p} - a_2\right) = \ln(a_3) + \frac{t}{4} \quad [\ln e = 1]$$

$$\Rightarrow \ln\left(\frac{50}{p} - a_2\right) = \frac{1}{4}t + \ln(a_3)$$

$$\therefore y = at + b$$

This is the linearized form

Where,

$$y = \ln\left(\frac{50}{p} - a_2\right)$$

$$a = \frac{1}{4}$$

$$b = \ln(a_3)$$

(ii)

$\Rightarrow$ We know:

The least square error as:

$$E(a, b) = \sum_{i=1}^n (at_i + b - y_i)^2$$

For minimum:

$$\frac{\partial E}{\partial a} = 0$$

4

$$\frac{\partial E}{\partial b} = 0$$

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

$$\Rightarrow \frac{\partial E}{\partial a} = 0$$

$$\Rightarrow \frac{\partial}{\partial a} \left[\sum_{i=1}^n (ax_i + b - y_i)^2 \right] = 0$$

$$\Rightarrow 2 \sum_{i=1}^n (ax_i + b - y_i) \times x_i = 0$$

$$\Rightarrow a \sum_{i=1}^n x_i^2 + b \sum_{i=1}^n x_i = \sum_{i=1}^n x_i y_i$$

①

$$\Rightarrow \frac{\partial E}{\partial b} = 0$$

$$\Rightarrow \frac{\partial}{\partial b} \left[\sum_{i=1}^n (ax_i + b - y_i)^2 \right] = 0$$

$$\Rightarrow 2 \sum_{i=1}^n (ax_i + b - y_i) \times 1 = 0$$

$$\Rightarrow a \sum_{i=1}^n x_i + b \sum_{i=1}^n 1 = \sum_{i=1}^n y_i$$

②

These are the Normal equations.

③

n	x_i	x_i^2	$y_i = \ln\left(\frac{50}{p} - a_2\right)$	$x_i y_i$
1	2	4	$y_1 = 0.363$	0.726
2	4	16	$y_2 = 0.641$	2.564
3	6	36	$y_3 = 0.805$	5.070
4	8	64	$y_4 = 1.100$	4.502
	$\sum_{i=1}^4 x_i = 20$	$\sum_{i=1}^4 x_i^2 = 120$	$\sum_{i=1}^4 y_i = 3.008$	$\sum_{i=1}^4 x_i y_i = 18.252$

n = number of data

Using these in eq ① & ② $\Rightarrow$

$$20a + 20b = 18.252 \quad \text{--- ③}$$

$$20a + 4b = 3.008 \quad \text{--- ④}$$

Now Solving the eq (iii) & (iv) we get:-

$$\therefore a = 0.188$$

$$b = 0.084$$

Also we know:-

$$b = \ln a_3$$

$$\Rightarrow \ln a_3 = 0.084$$

$$\therefore a_3 = e^{0.084}$$

$$a_3 = 1.087$$

Ans:-

(iv)

we have:-

$$P = \frac{50}{a_2 + a_3 e^{t/4}}$$

For $t = 10$:-

$$\therefore P = \frac{50}{1 + 1.087 e^{10/4}} = 3.510 \text{ lae}$$

$$\left. \begin{array}{l} a_2 = 1 \\ a_3 = 1.087 \\ t = 10 \end{array} \right\}$$

For Original Value:- $t = 0$:-

$$\therefore P = \frac{50}{1 + 1.087 e^{0/4}} = 23.05 \text{ lae}$$

Ans:-

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

(V)

MATLAB CODE:-

```
>> clc;
>> clear;
>> Xdata = [2, 4, 6, 8];
>> Ydata = [20.50, 17.25, 14.50, 11.58];
>> fun = @(Xdata, a) (a(1) ./ (a(2) + a(3) * EXP(Xdata/4)))
>> a(0) = [50, 1, 1.087];
>> a = lsqcurvefit(fun, a(0), Xdata, Ydata)
```

Example 2:-

⇒ Consider the data as

t	4	8	12	16	20
v	23.31	28.52	33.07	36.76	40.20

Using the equations: $V = 70 - ce^{-kt}$

(i) Find the value of c & k to 2 D.P

(ii) Find the time to the nearest Second, when the speed is 45 ms^{-1}

(iii) Use MATLAB function $a = \text{lsqcurvefit}(\text{fun}, a_0, X_{\text{data}}, Y_{\text{data}})$ for $V = 70 - ce^{-kt}$

(1)

⇒ Is given:-

$$V = 70 - ce^{-kt}$$

$$\Rightarrow ce^{-kt} = 70 - V$$

$$\Rightarrow \ln(ce^{-kt}) = \ln(70 - V)$$

$$\Rightarrow \ln(c) - kt = \ln(70 - V)$$

$$\Rightarrow \ln(70 - V) = (-k)t + \ln(c)$$

$$\therefore y = at + b$$

where,

$$y = \ln(70 - V)$$

$$a = -k$$

$$b = \ln(c)$$

Now we know:-

$$E(a, b) = \sum_{i=1}^n (at_i + b - y_i)^2$$

So, For minimum:-

$$\frac{\partial E}{\partial a} = 0$$

$$\Rightarrow \frac{\partial}{\partial a} \left[\sum_{i=1}^n (at_i + b - y_i)^2 \right] = 0$$

$$\Rightarrow 2 \sum_{i=1}^n (at_i + b - y_i) \times t_i = 0$$

$$\therefore a \sum_{i=1}^n t_i^2 + b \sum_{i=1}^n t_i = \sum_{i=1}^n t_i y_i$$

(1)

$$\frac{\partial E}{\partial b} = 0$$

$$\Rightarrow \frac{\partial}{\partial b} \left[\sum_{i=1}^n (at_i + b - y_i)^2 \right] = 0$$

$$\Rightarrow 2 \sum_{i=1}^n (at_i + b - y_i) \times 1 = 0$$

$$\therefore a \sum_{i=1}^n t_i + b \sum_{i=1}^n 1 = \sum_{i=1}^n y_i$$

(2)

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

n	t_i	V	$y_i = \ln(70 - V)$	t_i^2	$t_i y_i$
1	4	29.21	$y_1 = 3.84$	16	15.36
2	8	28.52	$y_2 = 3.72$	64	29.76
3	12	29.02	$y_3 = 3.60$	144	43.20
4	16	36.26	$y_4 = 3.40$	256	55.84
5	20	40.20	$y_5 = 3.30$	400	67.80
$\sum_{i=1}^5 t_i = 60$			$\sum_{i=1}^5 y_i = 18.04$	$\sum_{i=1}^5 t_i^2 = 880$	$\sum_{i=1}^5 t_i y_i = 211.96$

Using these in eq (i) & (ii) :-

$$880a + 60b = 211.96 \quad \text{--- (iii)}$$

$$60a + 5b = 18.04 \quad \text{--- (iv)}$$

Now Solving the eq (iii) & (iv) :-

$$\therefore a = -0.028$$

$$b = 3.04$$

Also we know,

$$a = -k$$

$$\therefore k = 0.028$$

$$b = \ln(c)$$

$$\therefore c = e^{3.04} = 51.41$$

Ans:-

(ii)

⇒ we have:—

$$V = 70 - ce^{-kt}$$

For $V = 45 \text{ m/s}$!—

$$45 = 70 - ce^{-kt}$$

$$\Rightarrow ce^{-kt} = 70 - 45$$

$$\Rightarrow e^{-kt} = \frac{70 - 45}{51.41}$$

$$\Rightarrow -kt = \ln(0.48)$$

$$\therefore t = \frac{\ln(0.48)}{-0.028}$$

$$= 26.215$$

Ans!—

(iii)

MATLAB CODE!—

» clc;

» clear;

» Xdata = [4.8, 12.16, 20];

» Ydata = [23.31, 28.52, 33.07, 36.96, 40.20];

» fun = @(Xdata, a) (70 - a(1) * EXP(-a(2) * Xdata));

» a(0) = [51.41, 0.028];

» a = ~~lsqcurvefit~~ lsqcurvefit(fun, a(0), Xdata, Ydata)

TOPIC NAME : _____

TIME : _____

DATE : / /

Numerical Differentiation

Ch 07

Numerical Differentiation! —

Formulas: —

⇒ First Derivatives: —

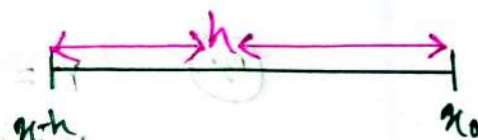
① 2-points Forward difference: —

$$f'(x) \approx \frac{f(x_0+h) - f(x_0)}{h}$$



② 2-points backward difference: —

$$f'(x) \approx \frac{f(x_0) - f(x_0-h)}{h}$$



③ 3-points central difference: —

$$f'(x) \approx \frac{f(x_0+h) - f(x_0-h)}{2h}$$



⇒ Second Derivatives: —

① 3-points Forward difference: —

$$f''(x) \approx \frac{f(x_0) - 2f(x_0+h) + f(x_0+2h)}{h^2}$$

② 3-points Backward difference: —

$$f''(x) \approx \frac{f(x_0) - 2f(x_0-h) + f(x_0-2h)}{h^2}$$

③ 3-points Central difference:—

$$f''(x_0) \approx \frac{f(x_0-h) - 2f(x_0) + f(x_0+h)}{h^2}$$

⇒ Richardson Extrapolation:—

① $M_R(x_0) = M(x_0, h_1) + \frac{M(x_0, h_1) - M(x_0, h_2)}{r^n - 1}$

② $r = \frac{h_2}{h_1}$

③ $n = 1$ for 2-points formula (Forward, Backward)
 $n = 2$ for central points formula.

¶ Example: 1:—

⇒ The values of Distance at various times are given below.

Time (t)	4	6	8	10	12
Distance (s)	7.58	12.02	18.38	26.42	36.40

① Using 3-points Central difference formula to estimate the speed at

(i) $t = 8$, (ii) $t = 7$, (iii) $t = 9$

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

⑥ Using two points formula and Extrapolation formula to estimate the speed at

(i) $t = 4$ (ii) $t = 12$

⑦ Use three-points Central difference formula & Extrapolation formula to estimate the speed at $t = 8$

⑧ Use three-points Central or forward or backward formula to estimate the acceleration at

(i) $t = 8$, (ii) $t = 4$, (iii) $t = 12$

⑨ Write down MATLAB code to estimate the speed at $t = 8$ using 3-points Central difference formula.

⑩ Use MATLAB functions " $sp = \text{spline}(x, y)$ ", " $dsp = \text{fnderc}(sp, \text{dorder})$ " & " $\text{fnval}(dsp, x_0)$ " to estimate the speed and acceleration at $t = 8.4$ & 12

⑪ we know: _____

⇒ Three points Central difference formula: _____

$$f'(x_0, h) = \frac{f(x_0+h) - f(x_0-h)}{2h} \quad \text{--- ①}$$

(i)

Is given:-

$$t = 8$$

$$h = (t_2 - t_1) = (8 - 6) = 2 \rightarrow \text{if } t = \text{even}$$

from eq (1):-

$$\begin{aligned}
 f'(8, 2) &= \frac{f(8+2) - f(8-2)}{2 \times h} \\
 &= \frac{f(10) - f(6)}{2 \times 2} \\
 &= \frac{26.42 - 12.02}{4} \\
 &= 3.5875
 \end{aligned}$$

(ii)

Is given:-

$$t = 7$$

$$h = \frac{t_2 - t_1}{2} = \frac{8 - 6}{2} = 1 \rightarrow \text{if } t = \text{odd}$$

from eq (1):-

$$\begin{aligned}
 f'(7, 1) &= \frac{f(7+1) - f(7-1)}{2 \times h} \\
 &= \frac{f(8) - f(6)}{2 \times 1} \\
 &= \frac{26.42 - 12.02}{2} \\
 &= 3.15
 \end{aligned}$$

(iii)

Is given—

$$t = 0$$

$$\therefore h = \frac{t_2 - t_1}{2} = \frac{10 - 8}{2} = 1$$

From eq (1) :-

$$f'(0,1) = \frac{f(0+1) - f(0-1)}{2 \times h}$$

$$= \frac{S(10) - S(8)}{2 \times 1}$$

$$= \frac{26.42 - 18.97}{2}$$

$$= 4.02$$

Ans.

(b)

we know—

① 2-points Forward formula—

$$f'(x_0, h) = \frac{f(x_0 + h) - f(x_0)}{h}$$

② 2-points Backward formula—

$$f'(x_0, h) = \frac{f(x_0) - f(x_0 - h)}{h}$$

③ Extrapolation formula—

$$M_p(x_0) = m(x_0, h_2) + \frac{m(x_0, h_2) - m(x_0, h_1)}{r^n - 1}$$

(i)

In given:-

$$t = 4$$

$$h_1 = t_2 - t_1 = 6 - 4 = 2$$

$$h_2 = t_3 - t_1 = 8 - 4 = 4$$

For $t = 4$; we must use forward difference formula:-

For h_1 :- $\therefore f(4, 2) = \frac{f(4+2) - f(4)}{h_1}$

$$v(4, 2) = \frac{s(6) - s(4)}{2}$$

$$= \frac{12.07 - 7.38}{2}$$

$$= 2.345$$

For h_2 :-

$$f(4, 4) = \frac{f(4+4) - f(4)}{h_2}$$

$$v(4, 4) = \frac{s(8) - s(4)}{4}$$

$$= \frac{18.37 - 7.38}{4}$$

$$= 2.7475$$

Now:- Extrapolated Value:-

$$V_P(4) = v(4, 2) + \frac{v(4, 2) - v(4, 4)}{\left(\frac{h_2}{h_1}\right)^2 - 1}$$

$$= 2.345 + \frac{2.345 - 2.7475}{\left(\frac{4}{2}\right)^2 - 1} = 1.0425$$

Ans:-

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

(ii)

In given:

$$t = 12$$

$$h_1 = t_2 - t_1 = 6 - 4 = 2$$

$$h_2 = t_2 - t_1 = 8 - 4 = 4$$

For $t = 12$: we must use Backward difference formula

For h_1 :

$$f'(12, 2) = \frac{f(12) - f(12 - h_1)}{h_1}$$

$$= \frac{S(12) - S(10)}{2}$$

$$= \frac{36.40 - 26.42}{2}$$

$$= 4.99$$

For h_2 :

$$f'(12, 4) = \frac{f(12) - f(12 - h_2)}{h_2}$$

$$= \frac{S(12) - S(8)}{4}$$

$$= \frac{36.40 - 18.37}{4} = 4.5075$$

Now:

Extrapolated value:

$$V_P(12) = V(12, 2) + \frac{V(12, 2) - V(12 - 4)}{\left(\frac{h_2}{h_1}\right)^2 - 1}$$

$$= 4.99 + \frac{4.99 - 4.5075}{\left(\frac{4}{2}\right)^2 - 1} = 5.4725$$

Ans:-

(3)

We know:-For speed:-

(i) 3-points central formula:-

$$f'(x_0, h) \approx \frac{f(x_0+h) - f(x_0-h)}{2h}$$

(ii) Extrapolation value:-

$$M_p(x_0) = m(x_0, h_1) + \frac{m(x_0, h_1) - m(x_0, h_2)}{r^n - 1}$$

Hence,

$$t = 4 + 4 = 8$$

$$h_1 = t_2 - t_1 = 6 - 4 = 2$$

$$h_2 = t_3 - t_1 = 8 - 4 = 4$$

For h_1 :-

$$f'(8, 2) \approx \frac{f(8+2) - f(8-2)}{2 \times h_1}$$

$$\therefore v(8, 2) = \frac{S(10) - S(6)}{2 \times 2} = \frac{26.42 - 22.07}{4} = 3.5875$$

For h_2 :-

$$f'(8, 4) \approx \frac{f(8+4) - f(8-4)}{2 \times h_2}$$

$$\therefore v(8, 4) \approx \frac{S(12) - S(4)}{2 \times 4} = \frac{36.40 - 7.38}{8} = 3.6275$$

Extrapolated value:-

$$M_p(8) = V_p(8) = v(8, 2) + \frac{v(8, 2) - v(8, 4)}{\left(\frac{h_2}{h_1}\right)^2 - 1}$$

$$\therefore V_P(8) = 3.5875 + \frac{3.5875 - 3.6275}{\left(\frac{4}{2}\right)^2 - 1}$$

$$= 3.5741$$

Ans:-

(d)

We know:-

(1) Three-point central formula:-

$$f''(x_0) \approx \frac{f(x_0-h) - 2f(x_0) + f(x_0+h)}{h^2}$$

(2) Three-points forward formula:-

$$f''(x_0) \approx \frac{f(x_0) - 2f(x_0+h) + f(x_0+2h)}{h^2}$$

(3) Three-points Backward formula:-

$$f''(x_0) \approx \frac{f(x_0) - 2f(x_0-h) + f(x_0-2h)}{h^2}$$

(1)

Is given:-

$$x = 8$$

$$\therefore h = \frac{1}{2} - \frac{1}{2} = 6 - 4 = 2$$

From eq (1) :-

$$f''(8,2) = \frac{f(8-2) - 2f(8) + f(8+2)}{2^2}$$

$$\therefore a(8,2) = \frac{S(6) - 2 \times S(8) + S(10)}{4}$$

$$= \frac{12.07 - 2 \times 18.92 + 26.42}{4}$$

$$= 0.4375$$

Ans:-

(ii)

we have:—

$$x = 4$$

$$\therefore h = x_2 - x_1 = 6 - 4 = 2$$

From eq (ii) $\Rightarrow$

$$f''(4,2) = \frac{f(4) - 2f(4+2) + f(4+2 \times 2)}{h^2}$$

$$\therefore a(4,2) = \frac{7.38 - 2 \times 12.07 + 18.37}{4}$$

$$= 0.4025$$

Ans:-

(iii)

we have:—

$$x = 12$$

$$h = x_2 - x_1 = 6 - 4 = 2$$

From eq (iii) $\Rightarrow$

$$f''(12,2) = \frac{f(12) - 2f(12-2) + f(12-2 \times 2)}{2^2}$$

$$a(4,2) = \frac{S(12) - 2S(10) + S(8)}{4} = \frac{26.40 - 2 \times 26.42 + 18.92}{4}$$

$$= 0.4825$$

Ans:-

③

MATLAB CODE:-

```

>>clc;
>>clear;
>> x=[6,8,10];
>> y=[12.07, 18.37, 26.42];
>> h= x(2)-x(1);
>> D1=(y(3)-y(1))/(2*h);
>> D2=(y(1)-2*y(2)+y(3))/h^2;

```

I

MATLAB CODE:-

```

>>clc;
>>clear;
>> x=[4,6,8,10,12];
>> y=[2.38, 12.07, 18.37, 26.42, 36.42];
>> sp=spline(x,y);
>> d1sp=fnder(sp,1);
>> ValD1=fval(d1sp, [8.4, 11]);
>> d2sp=fnder(sp,2);
>> ValD2=fval(d2sp, [8.4, 11]);

```

NUMERICAL INTEGRATION

Choi

Numerical Integration: —Formula: —⇒ Trapezoidal Rule for $N=1$ —

$$1. \int_{x_0}^{x_1} f(x) \cdot dx \approx \frac{h}{2} [f(x_0) + f(x_1)],$$

$$2. h = x_1 - x_0$$

⇒ Composite Trapezoidal Rule: —

$$1. \int_{x_0}^{x_n} f(x) \cdot dx \approx \frac{h}{2} [f(x_0) + 2\{f(x_1) + f(x_2) + \dots + f(x_{n-1})\} + f(x_n)]$$

$$2. h = \frac{x_n - x_0}{N}$$

⇒ Simpson's rule (Simpson $\frac{1}{3}$ rule) for $N=2$ —

$$1. \int_{x_0}^{x_2} f(x) \cdot dx \approx \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)]$$

$$2. h = \frac{x_2 - x_0}{N}$$

⇒ Composite Simpson's Rule: —

$$1. \int_{x_0}^{x_n} f(x) \cdot dx \approx \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + 2f(x_4) + \dots + 4f(x_{n-1}) + f(x_n)]$$

$$2. h = \frac{x_n - x_0}{N}$$

Richardson Extrapolation Formula:-① Trapezoidal Rule:-

$$\underline{\underline{1.}} \quad I_T = I_T(h_1) + \frac{I_T(h_1) - I_T(h_2)}{2^n - 1}$$

$$\underline{\underline{2.}} \quad R = \frac{h_2}{h_1} ; n = 2$$

② Simpson's Rule:-

$$\underline{\underline{1.}} \quad I_P = I_S(h_1) + \frac{I_S(h_1) - I_S(h_2)}{2^n - 1}$$

$$\underline{\underline{2.}} \quad R = \frac{h_2}{h_1} ; n = 4$$

Romberg Integration Formula:-

⇒ Romberg integration is an extrapolation formula of the trapezoidal rule for integration. The first order Richardson extrapolation values—

⇒ 1st order:-

$$I^{(1)}(2n, \frac{h}{2}) = I^{(0)}(2n, \frac{h}{2}) + \frac{I^{(0)}(2n, \frac{h}{2}) - I^{(0)}(n, h)}{2^2 - 1}$$

⇒ 2nd order:-

$$I^{(2)}(4n, \frac{h}{4}) = I^{(1)}(4n, \frac{h}{4}) + \frac{I^{(1)}(4n, \frac{h}{4}) - I^{(1)}(2n, \frac{h}{2})}{2^4 - 1}$$

TOPIC NAME : _____

DAY : _____

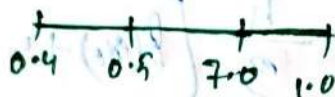
TIME : _____ DATE : / /

Example: 1:—

⇒ Evaluate $\int_{0.4}^{1.0} f(x) \cdot dx$ Numerically using the values given below.

x	0.4	0.5	0.7	1.0
$f(x)$	1.083	1.133	1.282	1.649

⇒ we have:—



Since the step size are not equal we can apply Trapezoidal rule for each subintervals.

we know:—

$$\Rightarrow \int_{x_0}^{x_1} f(x) \cdot dx = \frac{h}{2} [f(x_0) + f(x_1)]$$

For Each:—

$$\therefore \int_{0.4}^{1.0} f(x) \cdot dx = \frac{0.5-0.4}{2} [f(0.4) + f(0.5)] + \frac{0.7-0.5}{2} [f(0.7) + f(0.5)] + \frac{1.0-0.7}{2} [f(1.0) + f(0.7)]$$

$$= \frac{0.1}{2} [(1.083 + 1.133)] + \frac{0.2}{2} [(1.282 + 1.133)] + \frac{0.3}{2} [(1.282 + 1.649)]$$

$$= 0.7032$$

Ans:—

Example: 2:-

⇒ Evaluate $\int_{0.4}^{1.0} f(x) \cdot dx$ using Simpson's $\frac{1}{3}$ rule.

x	0.4	0.5	0.6	0.8	1.0
$f(x)$	1.083	1.133	1.197	1.377	1.649

⇒ We know:-

$$\Rightarrow \int_{x_0}^{x_2} f(x) \cdot dx = \frac{h}{3} [f(x_0) + 4f(x_1) + f(x_2)] ; h = \frac{x_2 - x_0}{2}$$

$$\therefore \Rightarrow \int_{0.4}^{1.0} f(x) \cdot dx = \int_{0.4}^{0.6} f(x) \cdot dx + \int_{0.6}^{1.0} f(x) \cdot dx$$

$$\therefore \int_{0.4}^{1.0} f(x) \cdot dx = \frac{(0.6-0.4)/2}{3} [f(0.4) + 4f(0.5) + f(0.6)] + \frac{(1.0-0.6)/2}{3} [f(0.6) + 4f(0.8) + f(1.0)]$$

$$= \frac{0.1}{3} [1.083 + 4 \times 1.133 + 1.197] + \frac{0.2}{3} [1.197 + 4 \times 1.377 + 1.649]$$

$$= 0.784$$

Ans:-

TOPIC NAME : _____

DAY : _____

TIME : _____ DATE : / /

Example: 3: _____

⇒ Evaluate $\int_{1.2}^{2.0} f(x) \cdot dx$

x	1.2	1.4	1.6	1.8	2.0
$f(x)$	3.728	4.124	4.525	5.123	5.626

(i) Use Trapezoidal rule and Richardson Extrapolation

(ii) Use Simpson's rule and Richardson Extrapolation.

(i)

⇒ we know—

$$\Rightarrow \int_{a_0}^{a_n} f(x) \cdot dx = \frac{h}{2} [f(x_0) + 2\{f(x_2) + f(x_3) + \dots + f(x_{n-1})\} + f(x_n)] \quad \text{--- (1)}$$

$$\Rightarrow \underline{I_p} = I_T(h_1) + \frac{I_T(h_1) - I_T(h_2)}{2^n - 1} \quad \text{--- (ii)}$$

For $h_1 = 0.2$

$$I_T(0.2) = \int_{1.2}^{2.0} f(x) \cdot dx$$

$$= \frac{h}{2} [f(1.2) + 2\{f(1.4) + f(1.6) + f(1.8)\} + f(2.0)]$$

$$= \frac{(2-1.2)}{2} [3.728 + 2\{4.124 + 4.525 + 5.123\} + 5.626]$$

$$= \frac{0.2}{2} \times [36.808] = 3.6808$$

For $h_2 = 0.4$:-

$$I_T(0.4) = \int_{1.2}^{2.0} f(x) \cdot dx$$

$$= \frac{(x_1 - x_0)N}{2} [f(x_1) + 2f(x_2) + f(x_3)]$$

$$= \frac{(2 - 1.2)/2}{2} [3.728 + 2 \times 4.525 + 5.626]$$

$$= 3.6808$$

Now put the value of in eq. (1) :-

$$I_T = I_T(0.2) + \frac{I_T(0.2) - I_T(0.4)}{\left(\frac{h_2}{h_1}\right)^2 - 1}$$

$$\left. \begin{aligned} \tau &= \frac{h_2}{h_1} \\ n &= 2 \text{ for Trapezoidal rule} \end{aligned} \right\}$$

$$= 3.6898 + \frac{3.6898 - 3.6808}{\left(\frac{0.4}{0.2}\right)^2 - 1}$$

$$= 3.6928$$

Ans:-

(ii)

we know-

$$\Rightarrow \int_{x_0}^{x_n} f(x) \cdot dx = \frac{h}{3} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)] \quad \text{--- (i)}$$

$$\Rightarrow I_R = I_S(h_1) + \frac{I_S(h_2) - I_S(h_1)}{(r^n) - 1} \quad \text{--- (ii)}$$

For $h_1 = 0.2$:-

$$I_S(0.2) = \int_{1.2}^{2.0} f(x) \cdot dx$$

$$= \frac{h}{3} [f(1.2) + 4f(1.4) + 2f(1.6) + 4f(1.8) + f(2.0)]$$

$$\therefore I_S(0.2) = \frac{0.2}{3} [3.728 + 4 \times 4.124 + 2 \times 4.525 + 4 \times 5.123 + 5.626]$$

$$= 3.6928$$

For $h_2 = 0.4$:-

$$I_S(0.4) = \int_{1.2}^{2.0} f(x) \cdot dx$$

$$= \frac{h}{3} [f(1.2) + 4f(1.6) + f(2.0)]$$

$$= \frac{0.4}{3} [3.728 + 4 \times 4.525 + 5.626]$$

$$= 3.6667$$

Put the value in eq (ii) =

$$\therefore I_R = 3.6928 + \frac{3.6928 - 3.6667}{\left(\frac{0.4}{0.2}\right)^4 - 1}$$

$$= 3.6940$$

Ans:-

$$\left\{ \begin{array}{l} n = 4 \\ r = \frac{h_2}{h_1} \end{array} \right.$$

Example: 4:—
mon
 ⇒ Given:—
mon

x	1.0	1.2	1.4	1.6	1.8
$f(x)$	1.831	2.892	3.515	4.643	5.926

→ Evaluate $\int_{1.0}^{1.8} f(x) \cdot dx$ by using Trapezoidal rule with 1.2 & 4 subintervals. Improve your result using Romberg Integration.

⇒ we have, using Trapezoidal rule:—
mon o mon o mon o mon

For $n=1$, $h = 1.8 - 1.0 = 0.8$:—
mon o mon o mon o mon

$$I^{(1)}(1, 0.8) = \frac{0.8}{2} [f(1.0) + f(1.8)]$$

$$= \frac{0.8}{2} [1.831 + 5.926] = 3.1028$$

For $n=2$, $h = 1.4 - 1.0 = 0.4$:—
mon o mon o mon o mon

$$I^{(2)}(2, 0.4) = \frac{h}{2} [f(1.0) + 2 \times f(1.4) + f(1.8)]$$

$$= \frac{0.4}{2} [1.831 + 2 \times 3.515 + 5.926]$$

$$= 2.9574$$

For $n=4$, $h=1.2-1=0.2$ —

$$I^{(4)}(4, 0.2) = \frac{h}{2} [f(1.0) + f(1.2) + f(1.4) + f(1.6)] \times 2 + f(1.8)$$

$$\therefore = \frac{0.2}{2} [1.891 + 2 \times (2.592 + 3.615 + 4.643) + 5.926]$$

$$= 2.9257$$

We know,

$\Rightarrow$ The 1st order Extrapolation rules: —

$$I^{(4)}(2n, \frac{h}{2}) = I^{(4)}(2n, \frac{h}{2}) + \frac{I^{(4)}(2n, \frac{h}{2}) - I^{(4)}(n, h)}{2^2 - 1}$$

For $n=1$, $h=0.8$ —

$$\therefore I^{(4)}(2 \times 1, \frac{0.8}{2}) = I^{(4)}(2 \times 1, \frac{0.8}{2}) + \frac{I^{(4)}(2, 0.4) - I^{(4)}(1, 0.8)}{4 - 1}$$

$$\therefore I^{(4)}(2, 0.4) = I^{(4)}(2, 0.4) + \frac{I^{(4)}(2, 0.4) - I^{(4)}(1, 0.8)}{4 - 1}$$

$$= 2.92574 + \frac{2.92574 - 2.1028}{3}$$

$$= 2.9089$$

For $n=2$, $h=0.4$ —

$$I^{(4)}(4, 0.2) = I^{(4)}(4, 0.2) + \frac{I^{(4)}(4, 0.2) - I^{(4)}(2, 0.4)}{4 - 1}$$

$$= 2.9257 + \frac{2.9257 + 2.92574}{3}$$

$$= 2.9151$$

Now,2nd order Extrapolation rules:-

$$I^{(2)}(4n, \frac{h}{4}) \approx I^{(2)}(4n, \frac{h}{4}) + \frac{I^{(2)}(4n, \frac{h}{4}) - I^{(2)}(2n, \frac{h}{2})}{2^2 - 1}$$

For $n=1, h=0.8$:-

$$\therefore I^{(2)}(4, 0.2) = I^{(2)}(4, 0.2) + \frac{I^{(2)}(4, 0.2) - I^{(2)}(2, 0.4)}{2^2 - 1}$$

$$= 2.9151 + \frac{2.9151 - 2.9089}{2^2 - 1}$$

$$= 2.9155$$

Ans:-

ORDINARY DIFFERENTIAL
EQUATIONS

Chaos

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Ordinary differential Equation:-

⇒ Consider the IVP:-

$$\frac{dy}{dx} = f(x, y),$$

And:- $y(x_0) = y_0.$

⇒ Euler Method:-

① $y_1 = y_0 + hf(x_0, y_0)$

② $y_2 = y_1 + hf(x_1, y_1)$

③ $y_{n+1} = y_n + hf(x_n, y_n)$

⇒ Second Order Runge-Kutta method (pk-2 method):-

$$y_1 = y_0 + \frac{1}{2} [k_1 + k_2]$$

Where,
among

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf(x_0 + h, y_0 + k_1)$$

⇒ Fourth Order Runge-Kutta Method (RK-4 method)

$$y_1 = y_0 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

Where,

$$k_1 = h f(x_0, y_0)$$

$$k_2 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right)$$

$$k_3 = h f\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right)$$

$$k_4 = h f(x_0 + h, y_0 + k_3)$$

† Example 1:-

⇒ Given that $y' = 2xy^2 - y$, where $y = 1$ at $x = 0$

- Estimate the value of $y(0.2)$ using Euler method with step size $h = 0.1$.
- Estimate $y(0.2)$ using RK-2 method.
- Estimate $y(0.4)$ using RK-4 method.
- Use 3-points Central difference formula to derive a recurrence relation for the above IVP.

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

(ii) Estimate the value of y at $x=0.4$ and 0.6 using this recurrence relation.

(c) Use MATLAB function " $[x_1, y_1] = \text{ode23}(f, [x_0, x_n], y_0)$ "

(i) To estimate the values of y in $0 \leq x \leq 1.4$ using $h=0.2$

(ii) To plot the curve in $[0, 1.4]$

(a)

⇒ Is given —

$$f(x, y) = 2xy^2 - y$$

We know —

$$y_1 = y_0 + h f(x_0, y_0)$$

$$\begin{aligned} \therefore y_1(0.1) &= 1 + 0.1 \times f(0, 1) \\ &= 1 + 0.1 \times -1 \\ &= 1 - 0.1 \\ &= 0.9 \end{aligned}$$

$$\begin{aligned} \therefore y_2(0.2) &= y_1 + h f(x_1, y_1) \\ &= 0.9 + 0.1 \times f(0.1, 0.9) \\ &= 0.9 + 0.1 \times -0.738 \\ &= 0.8262 \end{aligned}$$

here,

$$y_0 = 1$$

$$x_0 = 0$$

$$h = 0.1$$

$$\begin{aligned} f(0, 1) &= 2 \times 0 \times 1^2 - 1 \\ &= -1 \end{aligned}$$

$$y_1 = 0.9$$

$$x_1 = 0.1$$

$$f(0.1, 0.9) = -0.738$$

We know:-

$$y_1 = y_0 + \frac{1}{2} [k_1 + k_2] \quad \text{--- (i)}$$

Where,

$$k_1 = hf(x_0, y_0) \quad \text{--- (ii)}$$

$$k_2 = hf(x_0 + h, y_0 + k_1) \quad \text{--- (iii)}$$

Is given:-

$$h = 0.2$$

Now, From eq (ii) $\Rightarrow$:-

$$\begin{aligned} k_1 &= 0.2 \times f(0, 1) \\ &= 0.2 \times \{2 \times 0 \times 1^2 - 1\} \\ &= -0.2 \end{aligned}$$

here,

$$\begin{aligned} h &= 0.2 \\ x_0 &= 0 \\ y_0 &= 1 \end{aligned}$$

From eq (iii) $\Rightarrow$:-

$$\begin{aligned} k_2 &= 0.2 \times f(0 + 0.2, (1 - 0.2)) \\ &= 0.2 \times f(0.2, 0.8) \\ &= 0.2 \times \{2 \times 0.2 \times (0.8)^2 - 0.8\} \\ &= -0.1088 \end{aligned}$$

put the value of k_1 & k_2 in eq (i) $\Rightarrow$

$$\begin{aligned} \therefore y_1 &= y_0 + \frac{1}{2} [k_1 + k_2] \\ &= 1 + \frac{1}{2} [-0.2 - 0.1088] \\ &= 0.8456 \end{aligned}$$

Ans:-

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

We know:-

$$y_1 = y_0 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \quad \text{--- (i)}$$

Where,

$$k_1 = hf(x_0, y_0) \quad \text{--- (ii)}$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \quad \text{--- (iii)}$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) \quad \text{--- (iv)}$$

$$k_4 = hf(x_0 + h, y_0 + k_3) \quad \text{--- (v)}$$

Is given:-

$$h = 0.4$$

From eq (ii):-

$$k_1 = 0.4 \times f(0, 1)$$

$$= 0.4 \times \{2 \times 0 \times 1^2 - 1\}$$

$$= -0.4$$

here,

$$x_0 = 0$$

$$y_0 = 1$$

From eq (iii) =>

$$k_2 = 0.4 \times f\left(0 + \frac{0.4}{2}, 1 + \frac{-0.4}{2}\right)$$

$$= 0.4 \times f(0.2, 0.8)$$

$$= -0.2176$$

$$x_0 = 0$$

$$h = 0.4$$

$$k_1 = -0.4$$

$$y_0 = 1$$

From eq (iv) =>

$$k_3 = 0.4 \times f\left(0 + \frac{0.4}{2}, 1 + \frac{-0.2176}{2}\right)$$

$$= 0.4 \times f(0.2, 0.8912)$$

$$x_0 = 0$$

$$h = 0.4$$

$$k_2 = -0.2176$$

$$y_0 = 1$$

$$= 0.4 \times \frac{1}{2} \times 0.2 \times (0.8212)^2 = 0.8212$$

$$= -0.2294$$

From eq (v) :-

$$k_4 = 0.4 \times f(0 + 0.4, (1 - 0.2294))$$

$$= 0.4 \times f(0.4, 0.7706)$$

$$= 0.4 \times \frac{1}{2} \times 0.4 \times (0.7706)^2 = 0.7706$$

$$= -0.1182$$

Now, put all the value in eq (i) =

$$\therefore y_1 = y_0 + \frac{1}{6} [k_1 + 2(k_2 + k_3) + k_4]$$

$$= 1 + \frac{1}{6} [-0.4 + 2(-0.2176 - 0.2294) - 0.1182]$$

$$= 0.7646$$

Ans:-

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

(1)

(1)

⇒ We know:

⇒ Three point Central difference formula:

$$y'_n \approx \frac{y_{n+1} - y_{n-1}}{2h} \quad (1)$$

Is given:

$$y' = 2xy^2 - y$$

$$\therefore y'_n = 2x_n y_n^2 - y_n \quad (2)$$

From eq (1) & (2) ⇒

$$2x_n y_n^2 - y_n = \frac{y_{n+1} - y_{n-1}}{2h}$$

$$\therefore y_{n+1} = y_{n-1} + 2h[2x_n y_n^2 - y_n] \quad (3)$$

For $n=1$:

$$y_{2+1} = y_{1-1} + 2 \times h[2 \times x_1 y_1^2 - y_1]$$

$$\therefore y_2 = y_0 + 2h[2 \times x_1 y_1^2 - y_1]$$

$$= 1 + 2 \times 0.2[2 \times 0.2 \times (0.8456)^2 - 0.8456]$$

$$= 0.7761$$

here,

$$h = 0.2$$

$$x_1 = 0.2$$

$$x_0 = 0$$

$$y_0 = 1$$

$$y_1 = 0.8456$$

From (b)

For $n=2$:-
mon

$$y_{2+1} = y_{2-1} + 2h(2x_2 y_2^2 - y_2)$$

$$\therefore y_3 = y_1 + 2h(2x_2 y_2^2 - y_2)$$

$$= 0.8456 + 2 \times 0.2 [2 \times 0.4 \times (0.7761)^2 - 0.7761]$$

$$= 0.7070$$

here,

$$y_1 = 0.8456$$

$$y_2 = 0.7761$$

$$x_2 = 0.4$$

$$h = 0.2$$

$$\therefore x = 0.4 ; y = 0.7761$$

$$x = 0.6 ; y = 0.7070$$

Ans:-

(e)

(ii)

MATLAB CODE:-

>>clc;

>>clear;

>>f=@(x,y)(2*x*y^2-y);

>>[x1,y1]=ode23(f,[0,1.4],1)

>> solution=[x1,y1]

MATLAB CODE:-

>>clc;

>>clear;

>>f=@(x,y)(2*x*y^2-y);

>> x1n=0%0.28 1.4;

>>[x1,y1]=ode23(f,x1n,1)

>> plot(x1,y1)

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

System of Equations!

System of Equation:-

Formula:-

$$① \quad y' = f_1(t, y, z)$$

$$② \quad z' = f_2(t, y, z)$$

Where $y(t_0) = y_0$

$z(t_0) = z_0$

RK-2 for System:-

$$\Rightarrow y_1 = y_0 + \frac{1}{2} [k_1 + k_2]$$

Where

$$\therefore k_1 = hf_1(t_0, y_0, z_0)$$

$$k_2 = hf_1(t_0 + h, y_0 + k_1, z_0 + m_1)$$

$$\Rightarrow z_1 = z_0 + \frac{1}{2} [m_1 + m_2]$$

Where

$$m_1 = hf_2(t_0, y_0, z_0)$$

$$m_2 = hf_2(t_0 + h, y_0 + k_1, z_0 + m_1)$$

TOPIC NAME : _____

DAY : _____

TIME : _____

DATE : / /

Example 1:

Given the IVP:—

$$\frac{dy}{dx} = x + y^2 - z;$$

$$(y(1) = 2) \quad \text{①}$$

$$\frac{dz}{dx} = x^2 - 3y + z^2;$$

$$(z(1) = 2.5) \quad \text{②}$$

① Estimate $y(1.2)$ and $z(1.2)$ using RK-2 method

② Use MATLAB function " $[u_1, y_1] = \text{ode45}(\text{fun}, [x_0, x_1], y_0)$ "

(i) to estimate the value of y in $1 \leq x \leq 1.4$

(ii) to plot the solution curve in $[1, 1.4]$

We know:

$$y' = f_1(x, y, z); \quad y(x_0) = y_0$$

$$z' = f_2(x, y, z); \quad z(x_0) = z_0$$

Is given:

$$y' = f_1(x, y, z) = x + y^2 - z$$

$$z' = f_2(x, y, z) = x^2 - 3y + z^2$$

$$\therefore h = 0.2$$

Since,

$$x_0 = 1$$

$$y_0 = 2$$

$$z_0 = 2.5$$

$$x_1 = 1.2$$

$$\therefore h = x_1 - x_0 = 0.2$$

Again, we know:

$$y_1 = y_0 + \frac{1}{2} [k_1 + k_2] \quad \text{--- (i)}$$

$$z_1 = z_0 + \frac{1}{2} [m_1 + m_2] \quad \text{--- (ii)}$$

where,

$$k_1 = hf_1(x_0, y_0, z_0) \quad \text{--- (iii)}$$

$$m_1 = hf_2(x_0, y_0, z_0) \quad \text{--- (iv)}$$

$$k_2 = hf_1(x_0 + h, y_0 + k_1, z_0 + m_1) \quad \text{--- (v)}$$

$$m_2 = hf_2(x_0 + h, y_0 + k_1, z_0 + m_1) \quad \text{--- (vi)}$$

From eq (iii):

$$\begin{aligned} \therefore k_1 &= 0.2 \times f_1(1, 2, 2.5) \\ &= 0.2 \times (1 + 2^2 - 2.5) \\ &= 0.5 \end{aligned}$$

From eq (iv):

$$\begin{aligned} \therefore m_1 &= 0.2 f_2(1, 2, 2.5) \\ &= 0.2 \times (1^2 - 3 \times 2 + (2.5)^2) \\ &= 0.25 \end{aligned}$$

From eq (v):

$$\begin{aligned} k_2 &= 0.2 f_1(1.2, 2.25, 2.75) \\ &= 0.2 \times (1.2 + (2.5)^2 - 2.75) \\ &= 0.04 \end{aligned}$$

From eq (vi):

$$\begin{aligned} m_2 &= 0.2 f_2(1.2, (2.5), (2.75)) \\ &= 0.2 \times ((1.2)^2 - 3 \times 2.5 + (2.75)^2) = 0.30 \end{aligned}$$

put the value in eq (i) & (ii);—

$$\begin{aligned}\therefore y_1 &= y_0 + \frac{1}{2} [k_1 + k_2] \\ &= 2 + \frac{1}{2} [0.5 + 0.75] \\ &= \underline{2.725}\end{aligned}$$

$$\begin{aligned}z_1 &= z_0 + \frac{1}{2} [m_1 + m_2] \\ &= 2.5 + \frac{1}{2} [0.25 + 0.30] \\ &= \underline{2.775}\end{aligned}$$

Ans:—

(b)

(i)

Let

$$y_1 = y$$

$$y_2 = z$$

$\therefore$ The system becomes:

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} x + y_1^2 - y_2 \\ x^2 - 3y_1 + z^2 \end{pmatrix}; \quad \begin{pmatrix} y_1(1) \\ y_2(1) \end{pmatrix} = \begin{pmatrix} 2 \\ 2.5 \end{pmatrix}$$

MATLAB code:—

(i)

>> clear;

>> fun = @(x,y) (x + y(1)^2 - y(2); x^2 - 3*y(1) + y(2)^2);

>> [x1,y1] = ode45(fun, [1:0.1:1.4], [2, 2.5])

>> S = [x1, y1]

(ii)

>> plot(x1, y1)

Example 2:—
mon

⇒ Given the BVP:—
mon

$$y'' - y' = 1$$

$$y(0) = 1$$

$$y(1) = 2(e-1)$$

- Derive a recurrence relation using 3-points Central difference formulae with $h = 1/3$.
- Using the above relation solve the BVP.
- Express the equation as a system of first order equations.
- Use MATLAB function "Sol = bvp4c(odefun, bcfun, solinit)" to solve the BVP.
 - Estimate the values of y at $x = \frac{1}{3}, \frac{2}{3}, 0.4, 0.5$.
 - Plot the solution curve.

we know:—
mon

Three point Central Diff. formula:—
mon

$$f'_n = y'_n \approx \frac{y_{n+1} - y_{n-1}}{2h} \quad \text{--- (i)}$$

$$f''_n = y''_n \approx \frac{y_{n-1} - 2y_n + y_{n+1}}{h^2} \quad \text{--- (ii)}$$

is given—
mod

$$y'' - y' = 1 \quad \text{--- (1)}$$

put the value of y'' & y' in eq (1) $\Rightarrow$
o o o o o o o o o o

$$\therefore \frac{y_{n-1} - 2y_n + y_{n+1}}{h^2} - \frac{y_{n+1} - y_{n-1}}{2h} = 1$$

Taking $h = \frac{1}{2}$, we obtain:-
o o o o o o o o o o

$$9(y_{n+1} - 2y_n + y_{n-1}) - \frac{3}{2}(y_{n+1} - y_{n-1}) = 1$$

$$\Rightarrow (9 - \frac{3}{2})y_{n+1} - 18y_n + (9 + \frac{3}{2})y_{n-1} = 1$$

$$\Rightarrow \frac{15}{2}y_{n+1} - 18y_n + \frac{21}{2}y_{n-1} = 1$$

$$\Rightarrow 15y_{n+1} - 36y_n + 21y_{n-1} = 2 \quad \text{--- (11)}$$

(b)

In given:-
mod

The nodal points are,
o o o o o o o o o o

$$x_0 = 0,$$

$$x_1 = \frac{1}{3},$$

$$x_2 = \frac{2}{3}$$

$$x_3 = 1$$

$$y_0 = 1$$

$$y_3 = 2(e - 1)$$

For $n=1$:-

$$x_1 = \frac{1}{3}, y_0 = 1$$

put the value in eq (11) $\Rightarrow$

$$15y_2 - 36y_1 + 21y_0 = 2$$

$$\therefore -36y_1 + 15y_2 = 2 - 21 \times 1 = -19 \quad \text{--- (11)}$$

For $n=2$:-

$$x_2 = \frac{2}{3}, y_1 = 2(e-1)$$

put the value in eq (11) $\Rightarrow$

$$15y_3 - 36y_2 + 21y_1 = 2$$

$$\therefore 21y_1 - 36y_2 = 2 - 30(e-1) = -40.55 \quad \text{--- (12)}$$

Now, Solve the eq (11) & (12) :-

$$\therefore y_1 = 1.454$$

$$y_2 = 2.225$$

Let,

$$y_1 = y$$

$$y_2 = y' = y_1'$$

$$\therefore \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} y_2 \\ 1 + y_2 \end{pmatrix} ; \begin{pmatrix} y_1(0) \\ y_2(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 2(e-1) \end{pmatrix}$$

MATLAB CODE:-

>> clear;

>> clearc;

>> F = @(x,y)(y(2); 1+y(2));

>> be = @(ya,yb)(ya(1)-1; yb(1)-2*exp(1)+2);

>> xint = @(x)(1; 4);

>> solinit = bvpinit(linspace(0,1,3), [1, 2]);

>> sol = bvp4c(F, be, solinit);

(i) >> xint = [0, $\frac{1}{3}$, $\frac{2}{3}$, 1];

>> S_xint = deval(sol, xint)

(ii) >> xval = linspace(0,1);

>> yval = deval(sol, xval);

>> plot(xval, yval(1,:));

Sample Questions

1. Average price, P , of a certain type of second-hand car is believed to be related to its age, t years, by an equation of the form

$$P = \frac{50}{a + be^{\frac{t}{4}}}, \quad \text{where } a \text{ and } b \text{ are constants.}$$

Data from a recent newspaper give the following average price (in Taka) for used car of this type,

t (yrs)	2	4	6	8
P (lac)	20.50	17.25	14.50	11.75

- i) Estimate the values of a and b rounded to 3 significant figures.
 (ii) Estimate the value of a car of this type that is 10 years old and the original new price.

(iii) Use MATLAB function $a = \text{lsqcurvefit}(\text{fun}, a_0, \text{xdata}, \text{ydata})$ to fit the general form like $P = \frac{c}{a + be^{\frac{t}{4}}}$

2. The distance traveled by an object is given in the table below:

t (s)	8	9	10	11	12
$s(t)$ (m)	17.453	21.460	25.752	30.302	35.084

The speed and acceleration can be calculated by $v = \frac{ds}{dt}$ and $a = \frac{d^2s}{dt^2}$.

Using three-point central difference formula estimate the speeds at (i) $t = 9$, and (ii) $t = 10.5$.

- a) Using two-point formulas estimate the speeds at (i) $t = 8$, and (ii) $t = 12$.
 b) Use three points central difference formula and extrapolation to estimate speed at $t = 10$.
 c) Use three points central or forward or backward difference formula to estimate the accelerations at (i) $t = 10$, (ii) $t = 8$ and (iii) $t = 12$.
 d) Use MATLAB functions “**sp=spline(x,y)**”, “**fnder(sp, dorder)**” and “**fival(sp, xo)**” to estimate the speed and acceleration at time $t = 8.5, 10.5$, and 11.2 .
3. Evaluate $\int_0^1 \sqrt{1+2x^3} dx$ using Trapezoidal rule with 1, 2 and 4 subintervals.

Improve the results using **Romberg** integration.

4. Given the initial value problem

$$y' = 2x + x^2y \quad \text{with } y(0) = -1.$$

- a) Estimate the values of $y(0.2)$ using the Runge-Kutta method of order two.
 b) Estimate a value of $y(0.2)$ using the Runge-Kutta method of order four.
 c) (i) Use three point central difference formula for derivative to derive a recurrence relation for the above IVP.
 (ii) Estimate the values of y at $x = 0.4$ and 0.6 using this recurrence relation.
 d) Use MATLAB function “**[x1, y1]=ode45(f, [x0, xn], y0)**” (i) to estimate the values of y in $0 \leq x \leq 1.5$ using $h = 0.1$. (ii) to plot the solution curve in $[0, 1.5]$.

5. Consider the boundary value problem

$$y'' - 2y' + 2y = x, \quad y(0) = 1, \quad y(1) = 2.$$

- (a) Derive a recurrence relation for the above differential equation using three-point central difference formula for derivative with $h = 1/3$.
 (b) Using the above finite difference formula, solve the above boundary value problem.
 (c) Express the above initial value problem as a system of first order differential equations.
 (d) Use MATLAB function “**sol = bvp4c(odefun, bcfun, solinit)**” to solve the BVP.
 (i) Estimate the values of y at $x = 1/3$ and $2/3$.
 (ii) Plot the solution curve $y(x)$.

Ans. to the q: no: 1

(i)
is given:-
Now:-

$$p = \frac{50}{a_2 + a_3 e^{t/4}}$$

$$\Rightarrow a_2 + a_3 e^{t/4} = \frac{50}{p}$$

$$\Rightarrow a_3 e^{t/4} = \frac{50}{p} - a_2$$

$$\Rightarrow \ln(a_3 e^{t/4}) = \ln\left(\frac{50}{p} - a_2\right)$$

$$\Rightarrow \ln(a_3) + \ln e^{t/4} = \ln\left(\frac{50}{p} - a_2\right)$$

$$\Rightarrow \ln\left(\frac{50}{p} - a_2\right) = \frac{t}{4} + \ln(a_3)$$

This is the linearized form:-

Where,

$$y = \ln\left(\frac{50}{p} - a_2\right)$$

$$a = \frac{1}{4}$$

$$b = \ln(a_3)$$

Now, we know:-

The least square error as:-

$$E(a, b) = \sum_{i=1}^n (at_i + b - y_i)^2$$

Now, For minimum:-

$$\frac{\partial E}{\partial a} = 0$$

&

$$\frac{\partial E}{\partial b} = 0$$

Let
 $a = a_2$
 $b = a_3$
is given:-
$$p = \frac{50}{a + b e^{t/4}}$$

$$\Rightarrow \frac{\partial E}{\partial a} = 0$$

$$\Rightarrow \frac{\partial}{\partial a} \left[\sum_{i=1}^n (at_i + b - y_i)^2 \right] = 0$$

$$\Rightarrow a \sum_{i=1}^n t_i^2 + b \sum_{i=1}^n t_i = \sum_{i=1}^n t_i y_i$$

————— (I)

$$\Rightarrow \frac{\partial E}{\partial b} = 0$$

$$\Rightarrow \frac{\partial}{\partial b} \left[\sum_{i=1}^n (at_i + b - y_i)^2 \right] = 0$$

$$\Rightarrow a \sum_{i=1}^n t_i + nb = \sum_{i=1}^n y_i$$

————— (II)

n	t_i	t_i^2	$y_i = \ln\left(\frac{50}{P} - a_2\right)$	$t_i y_i$
1	2	4	$y_1 = 0.363$	0.726
2	4	16	$y_2 = 0.641$	2.564
3	6	36	$y_3 = 0.875$	5.370
4	8	64	$y_4 = 1.180$	9.442
	$\sum_{i=1}^4 t_i = 20$	$\sum_{i=1}^4 t_i^2 = 120$	$\sum_{i=1}^4 y_i = 3.079$	$\sum_{i=1}^4 t_i y_i = 18.102$

Now, From equation (I) & (II) $\Rightarrow$

$$120a + 20b = 18.102 \quad \text{————— (III)}$$

$$20a + 4b = 3.079 \quad \text{————— (IV)}$$

Now, Solving the eq (III) & (IV) we get:

$$a = 0.135$$

$$b = 0.003$$

Also we know:-

$$b = \ln(a_3)$$

$$\therefore a_3 = e^b$$

$$\Rightarrow a_3 = e^{0.003}$$

$$\Rightarrow a_3 = 1.007$$

$\therefore$ The value of $a = 1$
& The value of $b = 1.007$

Ans:-

(ii)

we have:-

$$P = \frac{50}{a + be^{t/4}}$$

For $t = 10$:-

$$\therefore P = \frac{50}{1 + 1.007e^{10/4}}$$

$$= 3.480 \text{ lac}$$

For $t = 0$:- the Original Value:-

$$\therefore P = \frac{50}{1 + 1.007e^{0/4}}$$

$$= 23.84 \text{ lac}$$

Ans:-

(iii)

MATLAB CODE:-

```
>> ecc;  
>> clear;  
>> Xdata = [2, 4, 6, 8];  
>> Ydata = [20.50, 17.25, 14.50, 11.75];  
>> fun = @(Xdata, a) (a(1) ./ (a(2) + a(3) * exp(Xdata/4)))  
>> a(0) = [50, 1, 1.007];  
>> a = lsqcurvefit(fun, a(0), Xdata, Ydata)
```

Ans: to the Q: no: 2

(i)

We know:-

⇒ Three point Central difference formula:-

$$f'(x_0, h) = \frac{f(x_0+h) - f(x_0-h)}{2h} \quad \text{--- (1)}$$

In given:-

$$t = 0, 10.5$$

$$h = 1, 0.5$$

Now! For t = 0:-

$$h = 1$$

∴ from eq (1) ⇒

$$f'(0, 1) = \frac{f(0+1) - f(0-1)}{2 \times 1}$$

$$\therefore f'(9,1) = \frac{f(10) - f(8)}{2 \times 1}$$

$$= \frac{25.752 - 17.459}{2}$$

$$= 4.1475$$

Now:- For $x = 10.5$ —
from
 $h = 0.5$

$\therefore$ from eq (1) $\Rightarrow$

$$f'(10.5; 0.5) = \frac{f(10.5+0.5) - f(10.5-0.5)}{2 \times 0.5}$$

$$= \frac{f(11) - f(10)}{2 \times 0.5}$$

$$= \frac{30.302 - 25.752}{1}$$

$$= 4.550$$

(a)

we know:-

1. 2-points Forward difference formula:-

$$f'(x) \approx \frac{f(x_0+h) - f(x_0)}{h} \quad \text{--- (1)}$$

② 2-points backward difference:—

$$f'(x) \approx \frac{f(x_0) - f(x_0 - h)}{h} \quad \text{--- ①}$$

Now:-

For $x=8$; $h=1$:—

From eq ① $\Rightarrow$

$$\begin{aligned} f'(8, 1) &= \frac{f(8+1) - f(8)}{1} \\ &= \frac{f(9) - f(8)}{1} \\ &= 21.460 - 17.453 \\ &= 4.007 \end{aligned}$$

Now:-

For $x=12$; $h=1$;

From eq ① $\Rightarrow$

$$\begin{aligned} f'(12, 1) &= \frac{f(12) - f(11)}{1} \\ &= \frac{35.084 - 30.302}{1} \\ &= 4.782 \end{aligned}$$

Ans:-

(b)

we know:-

① 3-points Central Difference formula:-

$$f'(x) \approx \frac{f(x_0+h) - f(x_0-h)}{2h} \quad \text{--- (1)}$$

② Richardson Extrapolation formula:-

$$M_p(x_0) = M(x_0, h_1) + \frac{M(x_0, h_1) - M(x_0, h_2)}{r^n - 1} \quad \text{--- (11)}$$

where $r = \frac{h_2}{h_1}$

$n=1$ for forward/backward

$n=2$ for Central D.P formula

Is given:-

$$t = 10$$

$$\therefore h_1 = 1; h_2 = 2$$

For $t=10$ & $h_1=1$:-

$$f'(10, 1) = \frac{f(11) - f(9)}{2 \times 1}$$

$$= \frac{30.302 - 21.460}{2} = 4.421$$

For $t=10$; $h_2=2$;

$$f'(10, 2) = \frac{f(12) - f(8)}{2 \times 2}$$

$$= \frac{35.084 - 17.453}{4} = 4.407$$

Now Put the values in eq (17) $\Rightarrow$

$$\begin{aligned}\therefore V_P(10) &= V(10,1) + \frac{V(10,1) - V(10,2)}{\left(\frac{h_2}{h_1}\right)^2 - 1} \\ &= 4.421 + \frac{4.421 - 4.407}{\left(\frac{2}{1}\right)^2 - 1} \\ &= 4.425\end{aligned}$$

Ans:-

(c)

We know:-

① 3-points Forward difference Formula:-

$$f''(x) \approx \frac{f(x_0) - 2f(x_0+h) + f(x_0+2h)}{h^2} \quad \text{--- (1)}$$

② 3-points Backward difference Formula:-

$$f''(x) \approx \frac{f(x_0) - 2f(x_0-h) + f(x_0-2h)}{h^2} \quad \text{--- (11)}$$

③ 3-points Central difference Formula:-

$$f''(x_0) \approx \frac{f(x_0-h) - 2f(x_0) + f(x_0+h)}{h^2} \quad \text{--- (111)}$$

For $t=10; h=1:-$

From eq (iii):

$$f''(10,1) = \frac{f(0) - 2f(10) + f(20)}{1^2}$$

$$= \frac{21.460 - 2 \times 25.752 + 30.302}{1}$$

$$= 0.258$$

For $t=8; h=1:-$

From eq (i):

$$f''(8,1) = \frac{f(8) - 2f(0) + f(10)}{1^2}$$

$$= 17.453 - 2 \times 21.460 + 25.752$$

$$= 0.285$$

For $t=12; h=1:-$

From eq (ii):

$$f''(12,1) = \frac{f(12) - 2f(11) + f(10)}{1^2}$$

$$= 35.084 - 2 \times 30.302 + 25.752$$

$$= 0.232$$

Ans:-

MATLAB CODE:-

(d)

```
>> clc;  
>> clear;  
>> x = [8, 10, 12, 14, 16];  
>> y = [17.453, 21.460, 25.752, 30.302, 35.084];  
>> sp = spline(x, y);  
>> d1sp = fnder(sp, 1);  
>> ValD1 = fnval(d1sp, [8.5, 10.5, 11.2]);  
>> d2sp = fnder(sp, 2);  
>> ValD2 = fnval(d2sp, [8.5, 10.5, 11.2]);
```

Ans:-

Ans! to the q! no! 3!:-

⇒ We have, using Trapezoidal Rule:-

$$f(x) = \sqrt{1 + 2x^3}$$

For $n=1$; $h = \frac{1-0}{1} = 1$:-

$$I^{(1)}(1, 1) = \frac{1}{2} [f(0) + f(1)]$$

$$= \frac{1}{2} [(1 + \sqrt{2})] = 1.207$$

For $n=2$ $h = \frac{1-0}{2} = \frac{1}{2}$:-

$$I^{(2)}(1, \frac{1}{2}) = \frac{0.5}{2} [f(0) + 2f(0.5) + f(1)]$$

$$= \frac{1}{4} [1 + 2 \times 1.118 + \sqrt{2}] = 1.242$$

For $n=4$; $h = \frac{1-0}{4} = \frac{1}{4}$:-

$$I^{(1)}(4, \frac{1}{4}) = \frac{1}{8} [f(0) + 2 \times \{f(0.25) + f(0.5) + f(0.75)\} + f(1)]$$

$$\approx \frac{1}{8} [1 + 2 \times \{1.015 + 1.118 + 1.357\} + \sqrt{3}]$$

$$= 1.214$$

We know:-

$\Rightarrow$ The 1st order Extrapolation rules:-

$$I^{(1)}(2n, \frac{h}{2}) = I^{(0)}(2n, \frac{h}{2}) + \frac{I^{(0)}(2n, \frac{h}{2}) - I^{(0)}(n, h)}{2^2 - 1}$$

Now, For $n=1$; $h=1$:-

$$I^{(1)}(2, 0.5) = I^{(0)}(2, 0.5) + \frac{I^{(0)}(2, 0.5) - I^{(0)}(1, 1)}{4 - 1}$$

$$= 1.242 + \frac{1.242 - 1.366}{3}$$

$$= 1.200$$

Now, For $n=2$; $h=0.5$:-

$$I^{(1)}(4, 0.25) = I^{(0)}(4, 0.25) + \frac{I^{(0)}(4, 0.25) - I^{(0)}(2, 0.5)}{4 - 1}$$

$$= 1.214 + \frac{1.214 - 1.242}{3}$$

$$= 1.204$$

Now

2nd Order Extrapolation rules:-

$$I^{(2)}(4n, \frac{h}{4}) \approx I^{(1)}(4n, \frac{h}{4}) + \frac{I^{(1)}(4n, \frac{h}{4}) - I^{(1)}(2n, \frac{h}{2})}{2^4 - 1}$$

For $n=1, h=1$:-

$$I^{(2)}(4, 0.25) = I^{(1)}(4, 0.25) + \frac{I^{(1)}(4, 0.25) - I^{(1)}(2, 0.5)}{2^4 - 1}$$

$$= 1.204 + \frac{1.204 - 1.200}{2^4 - 1}$$

$$= 1.204266$$

N	H	$O(h^2)$	$O(h^4)$	$O(h^6)$
1	1	1.366		
2	0.5	1.242	1.200	
4	0.25	1.214	1.204	1.204266

Ans:-

Ans. to the Q. no. 4:-

(a)

Is given,

$$y' = 2x + x^2y; y(0) = -1$$

we know:-

For R-K-2 Method:-

$$y_1 = y_0 + \frac{1}{2} [k_1 + k_2] \quad \text{--- (i)}$$

where, $k_1 = hf(x_0, y_0) \quad \text{--- (ii)}$

$$k_2 = hf(x_0 + h, y_0 + k_1) \quad \text{--- (iii)}$$

Now,

From eq (ii) $\Rightarrow$:-

$$\begin{aligned} k_1 &= 0.2 f(0, -1) \\ &= 0.2 \times \{2 \times 0 + 0^2 \times -1\} \\ &= 0 \end{aligned}$$

From eq (iii) $\Rightarrow$:-

$$\begin{aligned} k_2 &= 0.2 \times f(0.2, -1) \\ &= 0.2 \times [2 \times 0.2 + (0.2)^2 \times -1] \\ &= 0.072 \end{aligned}$$

put the value of k_1, k_2 in eq (i) $\Rightarrow$

$$\begin{aligned} \therefore y_1 &= -1 + \frac{1}{2} [0 + 0.072] \\ &= -0.964 \end{aligned}$$

Ans:-

herce,

$$h = 0.2$$

$$x_0 = 0$$

$$y_0 = -1$$

(6)

we know:-

RK-4 method:-

$$y_1 = y_0 + \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \quad \text{--- (I)}$$

where,

$$k_1 = hf(x_0, y_0) \quad \text{--- (II)}$$

$$k_2 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}\right) \quad \text{--- (III)}$$

$$k_3 = hf\left(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}\right) \quad \text{--- (IV)}$$

$$k_4 = hf(x_0 + h, y_0 + k_3) \quad \text{--- (V)}$$

Now:-

From eq (II):-

$$\begin{aligned} k_1 &= 0.2 \times f(0, -1) \\ &= 0.2 \times (2 \times 0 + 0^2 \times -1) \\ &= 0 \end{aligned}$$

From eq (III):-

$$\begin{aligned} k_2 &= 0.2 f(0.1, -1) \\ &= 0.2 \times [2 \times 0.1 + (0.1)^2 \times -1] \\ &= 0.038 \end{aligned}$$

From eq (IV):-

$$\begin{aligned} k_3 &= 0.2 f(0.1, -0.981) \\ &= 0.2 \times [2 \times 0.1 + (0.1)^2 \times (-0.981)] \\ &= 0.038038 \end{aligned}$$

here,

$$h = 0.2$$

$$y_0 = -1$$

$$x = 0$$

From eq (V):—

$$\begin{aligned}K_4 &= 0.2 \times \frac{1}{2} (0.22 - 0.961) \\&= 0.2 \times [2 \times 0.2 + (0.2)^2 \times (-0.961)] \\&= 0.0723\end{aligned}$$

put all the value in eq (I):

$$\begin{aligned}\therefore y_1 &= -1 + \frac{1}{6} [0 + 2 \times 0.038 + 2 \times 0.038038 + 0.0723] \\&= -0.962604\end{aligned}$$

Ans:—

(C)

(1)

we know:—

Three point Central difference Formula:—

$$y'_n \approx \frac{y_{n+1} - y_{n-1}}{2h} \quad \text{--- (1)}$$

Is given:—

$$y' = 2x + 2x^2$$

$$\therefore y'_n = 2x_n + 2x_n^2 \quad \text{--- (11)}$$

From eq (i) & (ii) $\Rightarrow$:-

$$2x_n + x_n^2 y_n = \frac{y_{n+1} - y_{n-1}}{2h}$$

$$\Rightarrow y_{n+1} = y_{n-1} + 2h[2x_n + x_n^2 y_n] \quad \text{--- (iii)}$$

This is the recurrence relation

Ans:-

(ii)

From eq (iii) $\Rightarrow$

For $n=1$:-

$$\begin{aligned} y_2 &= y_0 + 2h[2x_1 + x_1^2 y_1] \\ &= -1 + 2 \times 0.2[2 \times 0.2 + (0.2)^2 \times -0.962] \\ &= -0.855 \end{aligned}$$

here

$$\begin{aligned} x_1 &= 0.2 \\ y_0 &= -1 \\ x_0 &= 0 \\ h &= 0.2 \\ y_1 &= -0.962 \end{aligned}$$

For $n=2$:-

$$\begin{aligned} y_3 &= y_1 + 2h[2x_2 + x_2^2 y_2] \\ &= -0.962 + 2 \times 0.2[2 \times 0.4 + (0.4)^2 \times -0.855] \\ &= -0.696 \end{aligned}$$

$$\begin{aligned} x_2 &= 0.4 \\ h &= 0.2 \\ y_1 &= -0.962 \\ y_2 &= -0.855 \end{aligned}$$

Ans:-

①

②

MATLAB CODE:-

```

>>clc;
>>clear;
>>f=@(x,y)(2*x+x^2*y)
>>[x1,y1]=ode45(f,[0,1.5],-1)
>>Solution=[x1,y1]

```

①

```

>>xin=0:0.2:1.5;
>>[x2,y2]=ode45(f,xin,-1)
>>plot(x2,y2)

```

Ans:-

①

Ans: to the Q: no: 5:-

We know:-

The 3-points Central Diff. formula:-

$$y'_n \approx \frac{y_{n+1} - y_{n-1}}{2h} \quad \text{--- (1)}$$

$$y''_n \approx \frac{y_{n-1} - 2y_n + y_{n+1}}{h^2} \quad \text{--- (2)}$$

Now,

Is given:-

$$y'' - 2y' + 2y = x \quad \text{--- (i)}$$

Now put the value of eq (i) & (ii) in eq (iii)

$$\therefore \frac{y_{n-1} - 2y_n + y_{n+1}}{h^2} - 2 \times \frac{y_{n+1} - y_{n-1}}{2h} + 2y_n = x_n$$

Taking $h = \frac{1}{3}$,

$$\therefore 9(y_{n-1} - 2y_n + y_{n+1}) - 2 \times \frac{3}{2}(y_{n+1} - y_{n-1}) + 2y_n = x_n$$

$$\Rightarrow (9+3)(y_{n-1}) + (-18+2)y_n + (9-3)(y_{n+1}) = x_n$$

$$\Rightarrow 12y_{n-1} - 16y_n + 6y_{n+1} = x_n \quad \text{--- (iv)}$$

b

Is given:-

The nodal points are:-

$$\left. \begin{array}{l} x_0 = 0 \\ x_1 = \frac{1}{3} \\ x_2 = \frac{2}{3} \\ x_3 = 1 \end{array} \right\} \begin{array}{l} y_0 = 1 \\ y_3 = 2 \end{array}$$

For $n=1$:-

$$K_1 = \frac{1}{3}, Y_0 = 1$$

$\therefore$ put the value in eq (IV):-

$$12Y_0 - 16Y_1 + 6Y_2 = K_1$$

$$\Rightarrow -16Y_1 + 6Y_2 = K_1 - 12Y_0$$

$$\Rightarrow -16Y_1 + 6Y_2 = \frac{1}{3} - 12 \times 1 = -11.67 \quad \text{--- (V)}$$

For $n=2$:-

$$K_2 = \frac{2}{3}, Y_3 = 2$$

put the value in eq (IV):-

$$12Y_1 - 16Y_2 + 6Y_3 = K_2$$

$$\Rightarrow 12Y_1 - 16Y_2 = K_2 - 6Y_3$$

$$\Rightarrow 12Y_1 - 16Y_2 = \frac{2}{3} - 6 \times 2 = -11.33 \quad \text{--- (VI)}$$

Now, Solve the eq (V) & (VI):-

$$\therefore Y_1 = 1.384$$

$$Y_2 = 1.746$$

Ans:-

③

Let

$$y_1 = y; \quad x_1 = x$$

$$\therefore y_2 = y' = y_1'$$

$$\therefore \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} y_2 \\ x - 2y_1 + 2y_2 \end{pmatrix}; \quad \begin{pmatrix} y_1(0) \\ y_2(0) \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

④

MATLAB CODE:-

```
>>clc;
```

```
>>clear;
```

```
>>F=@(x,y)(y(2);x(1)-2*y(1)+2*y(2));
```

```
>>bc=@(ya,yb)(ya(1)-1;yb(1)-2);
```

```
>>yint=@(x)(1;4);
```

```
>>Solinit=bvpinit(linspace(0,1,10),(1,0));
```

```
>>Sol=bvp4c(F,bc,Solinit);
```

① >> xint=[0, 1/3, 2/3, 1];

```
>>S_xint=deval(Sol,xint)
```

② >> xval=linspace(0,1,100);

```
>> yval=deval(Sol,xval);
```

```
>> plot(xval,yval(1,:));
```